

MEDTECH VENTURE STUDIO

LED FIXATION DEVICE

OCU GUIDE

PHASE II ASSESSMENT

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OVERVIEW



DISEASE STATE

Ophthalmic conditions like cataracts and glaucoma require surgeries that depend on precise eye fixation.



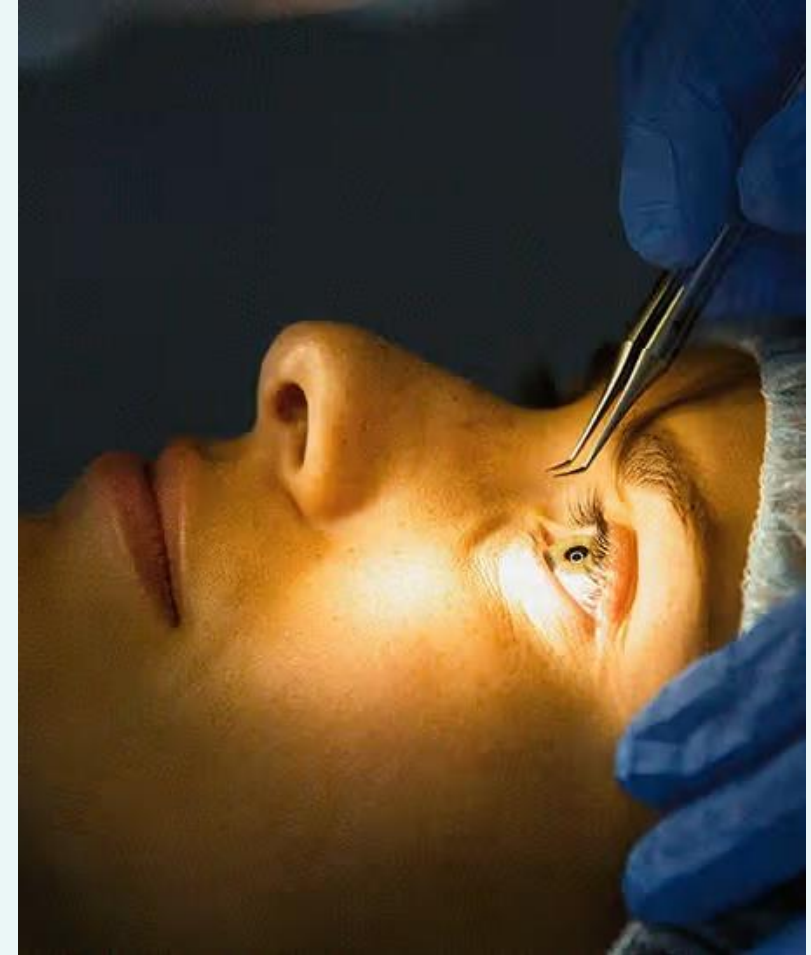
PROCEDURE

Ocu Guide improves surgical precision by using a customizable LED System that reduces verbal cues and manual eye manipulation — minimizing complications, time, and costs.



PROBLEM

Current fixation methods are inconsistent, leading to surgical delays, complications, and increased costs.



OVERVIEW



DISEASE STATE

4M+ cataract surgeries are performed annually in the U.S., with **20M+ globally**, driven by an aging population.



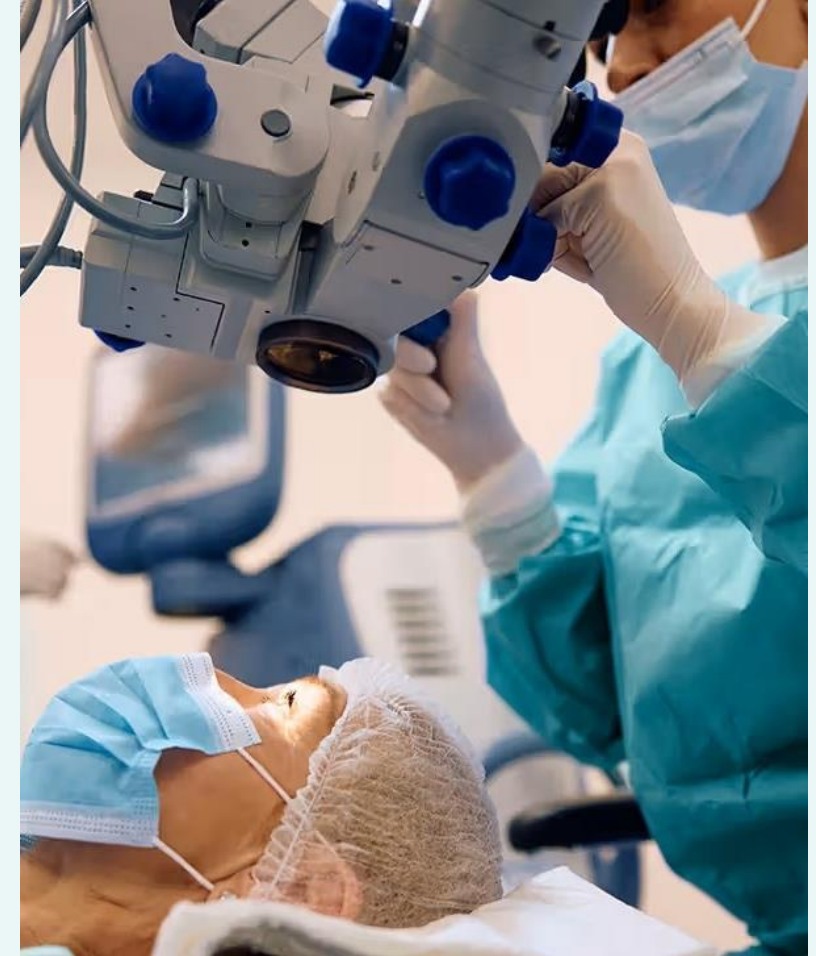
PROCEDURE

Up to **10 intraocular injections/day** per ophthalmologist—each requiring precise, steady fixation.



PROBLEM

One particular complication, called a posterior capsular tear, can increase the cost of surgery by an average of \$1,111.07 per case



PROBLEM CONFIRMATION INTERVIEW

1. **Eye movement is a consistent challenge** across retinal surgeries, intraocular injections, and general ophthalmic procedures.
2. **Manual fixation methods**—like handheld lights and head stabilization—are frequently used but **unreliable**, especially in longer procedures.
3. **Surgeons often pause mid-procedure** to readjust fixation due to patient movement, affecting flow and precision.
4. **Ophthalmologists performing frequent injections** reported **high variability in patient compliance**, increasing risk of error.
5. **Scrub nurses and OR assistants** noted that manual support tools **limit mobility** and **disrupt workflow**, requiring additional staff attention.
6. There is a **clear need for a consistent, hands-free, and surgeon-controlled fixation method** that improves efficiency and reduces reliance on patient cooperation or extra personnel.

This is based on interviews that the OcuGuide team conducted as we were not able to secure any interviews for this assessment.

PROPOSED SOLUTION

Circular LED Array Plate

- Custom-designed base with a central cutout fits around microscope lens
- 360° LED placement enables full-range eye guidance (up, down, left, right & diagonal)
- Multiple LEDs at varying angles allow for precise, customizable fixation points

Microscope-Mounted Design

- Attaches directly to surgical microscopes (e.g., Leica M844, Wild M690)
- Close proximity to the patient's eye ensures accuracy without obstructing surgeon workflow
- Compatible with most microscope models using customizable adapters

Adaptive LED Control

- LEDs controlled via external touchscreen interface
- Adjustable color, intensity, and flashing patterns to match surgical stages and patient needs
- Designed to improve fixation, reduce eye fatigue, and enhance surgical efficiency

Hands-Free, Workflow-Friendly

- Operated by staff—not the surgeon—via external control panel
- Avoids bulky ceiling mounts or disruptive stand-alone systems
- Enhances safety without compromising OR space or visibility

VALUE PROPOSITION

Enhanced Surgical Precision

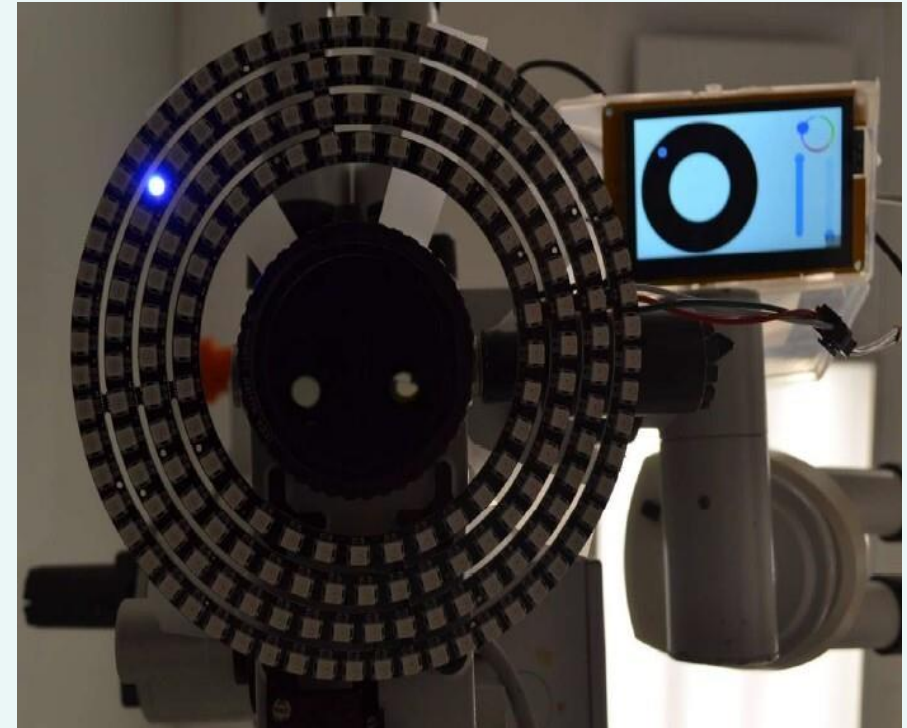
Provides consistent, hands-free eye fixation, improving accuracy in ophthalmic procedures.

Operational Efficiency

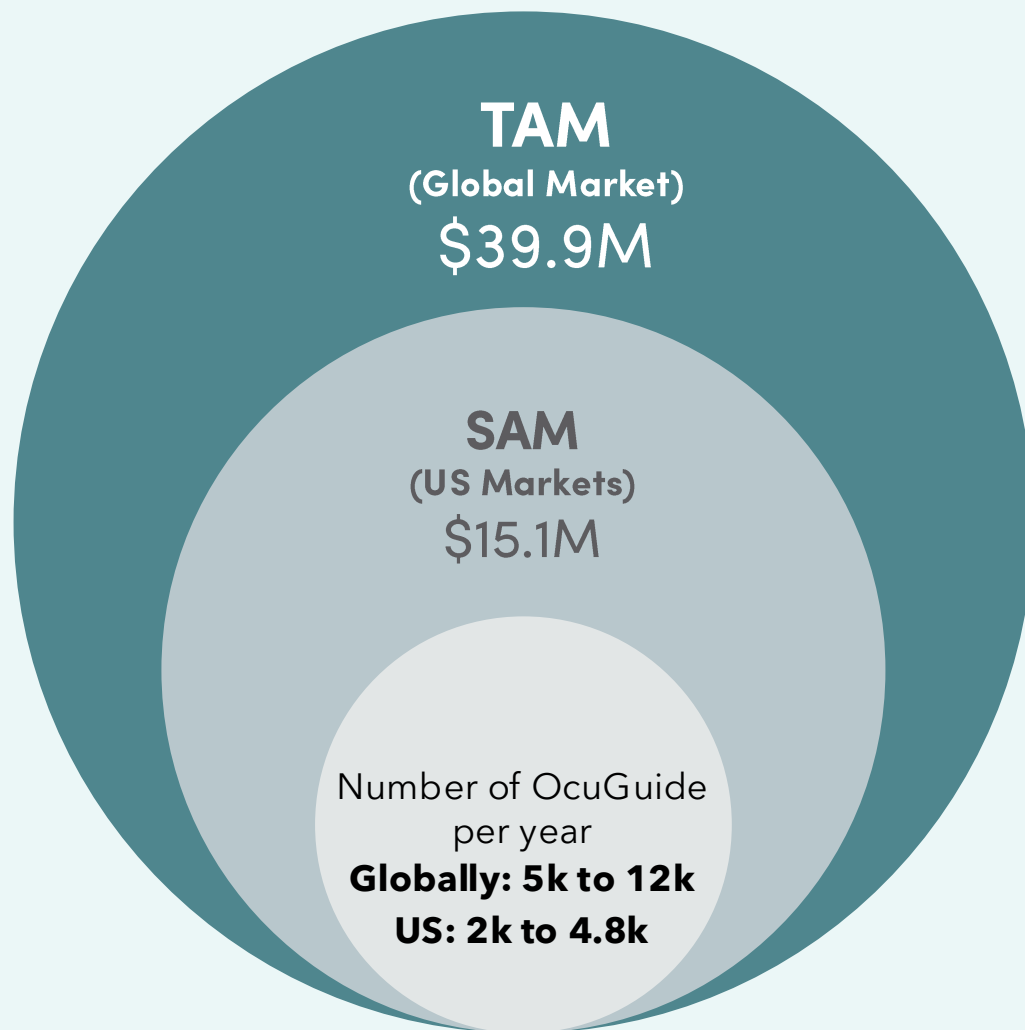
Minimizes the need for surgeon intervention, potentially reducing surgery duration and increasing throughput.

Complication Mitigation

Reduces fixation-related surgical errors, decreasing associated healthcare costs.



MARKET OVERVIEW



The global ophthalmic microscope market is valued at \$634 Million with the U.S. accounting 38% and growing at **4.7% CAGR**.

Estimated annual US unit sales: **6700** microscopes, based on average prices **\$50k-\$125k**

OcuGuide, priced at around \$4500, targets high-end surgical microscopes- commonly used in operating rooms for procedures like cataract and glaucoma surgeries.

As hospitals invest in advanced visualization tools, the market for OcuGuide would grow steadily.

OcuGuide's **integration with premium systems** aligns with rising demand for **precision, safety, and patient cooperation tools** during ophthalmic surgeries.

COMPETITIVE LANDSCAPE

Invasiveness ↓	Technique	Pros	Limitations	Complications
	Verbal Commands	Non-invasive, no cost	Relies on patient cooperation; language barriers	Poor fixation can lead to suboptimal outcomes
	External Light	Inexpensive, non-invasive	Requires additional personnel; limited effectiveness for central fixation	Poor fixation-related issues
	Head Stabilization	Non-invasive, moderately priced	Doesn't restrict eye movement	Ineffective for precise tasks
	Fixation Ring	Some surgeon control; no incision needed	Occupies surgeon's hand; discomfort; only useful during incision	Corneal abrasions, anatomy distortion
	Forceps	Strong control over the eye	Occupies one hand; painful; traumatic to tissue	Bleeding, trauma, anatomy distortion
	Traction Sutures	Provides firm control	Limited range of motion once applied; discomfort	Trauma, anatomy distortion
	Nerve Block	High control	Still needs manipulation; possible systemic side effects	Nerve damage, toxicity, hemorrhage
	General Anesthesia	Full control, no patient interference	Costly, risky, not suitable for all patients	Intubation risks, nausea, dental injury

COMPETITIVE ANALYSIS

Simple methods like verbal cues, lights, and head stabilization:

- Easy and affordable
- Highly dependent on patient cooperation
- Inconsistent, especially in longer surgeries

Intermediate methods such as fixation rings, forceps, and sutures:

- Offer better control
- Often uncomfortable for patients and interrupt surgical flow
- May cause tissue trauma or distort anatomy

Extreme measures like nerve blocks or general anesthesia:

- Provide complete control
- Costly, require significant prep, and not ideal for routine procedures

Current fixation methods fall into two extremes:

- **Low-cost, non-invasive** techniques with poor control
- **High-control methods** that are invasive, uncomfortable, or disruptive

Common drawbacks across all current options:

- Limited consistency, patient discomfort, inefficiency, and workflow disruption

OcuGuide addresses this gap by offering:

- Consistent fixation control
- Non-invasive design
- Seamless integration into workflow without extra burden on staff or patients

REGULATORY ASSESSMENT

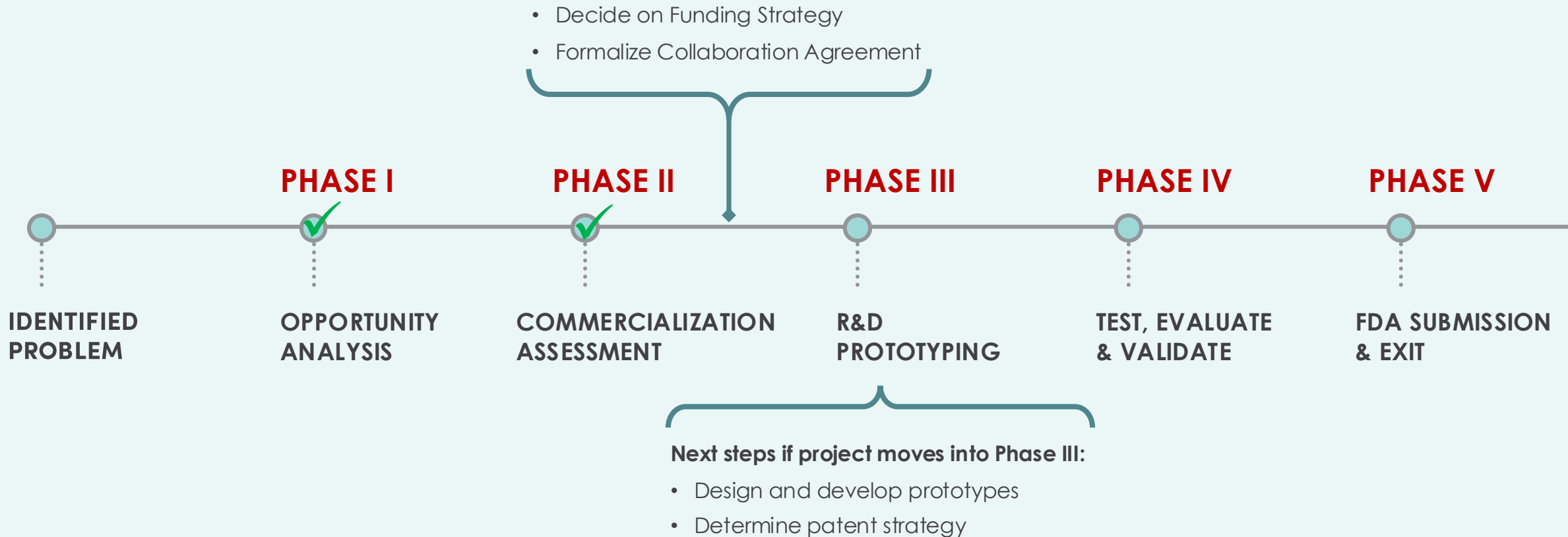
 **INTENDED SUBMISSION:** Letter to file

CLASSIFICATION INFORMATION					
FDA CFR No.	PRODUCT CODE	CLASS	REVIEW PANEL	PRODUCT CODE DESCRIPTION	SUBMISSION TYPE
886.4070	HLI	I	Ophthalmic	Microscope, Surgical, With Accessories	510(k) Exempt

REIMBURSEMENT & COVERAGE

CODING					
CODE	DESCRIPTION	WRVU	FACILITY RVU	ESTIMATED TRVU	PRICE
66984	Cataract extraction with IOL implant (extracapsular)	7.35	8.23	16.13	\$521.75
66982	Complex cataract surgery	10.25	11.05	22.09	\$780.63
67010	Vitrectomy, anterior segment	7.06	8.62	16.23	\$524.98
67036	Pars plana vitrectomy	12.13	13.46	26.53	\$858.15
67028	Intravitreal injection of a pharmacologic agent (separate procedure)	1.44	1.18	2.74	\$88.63

NEXT STEPS





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THANK YOU

innovation for a better tomorrow



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